

Test II: Automated Scoring

AutoEIT GSoC 2026 Evaluation

Automated meaning-based scoring of Spanish EIT transcriptions using the Ortega (2000) rubric.

```
In [13]: import pandas as pd
import openpyxl
from openpyxl import load_workbook
import re
import os
import unicodedata
from sentence_transformers import SentenceTransformer, util

SCORING_EXCEL_PATH = r"C:\Users\kashi\OneDrive\Desktop\AutoEIT datasets\AutoEIT
OUTPUT_SCORING_PATH = r"C:\Users\kashi\OneDrive\Desktop\AutoEIT\autoeit\output\t

os.makedirs(os.path.dirname(OUTPUT_SCORING_PATH), exist_ok=True)
print("Setup complete")
```

Setup complete

```
In [14]: print("Loading multilingual scoring model...")
model = SentenceTransformer("paraphrase-multilingual-MiniLM-L12-v2")
print("Model loaded!")
```

Loading multilingual scoring model...

Loading weights: 0% | 0/199 [00:00<?, ?it/s]

BertModel LOAD REPORT from: sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2

Key	Status	
-----+-----+--		
embeddings.position_ids	UNEXPECTED	

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.

Model loaded!

```
In [15]: def clean_stimulus(text):
        """Remove syllable count in parentheses from stimulus"""
        if not text:
            return ""
        return re.sub(r'\(\\d+\\)\s*$', '', str(text)).strip()

def clean_text(text):
    """Remove transcription notation before scoring"""
    if not text or not isinstance(text, str):
        return ""
    text = re.sub(r'\[.*?\]', '', text) # Remove [pause], [gibberish] etc
    text = re.sub(r'\bxxx\b', '', text, flags=re.IGNORECASE)
    text = re.sub(r'\s+', ' ', text).strip()
    return text
```

```

def normalize_text(text):
    """Remove accents for comparison"""
    return ''.join(
        c for c in unicodedata.normalize('NFD', text)
        if unicodedata.category(c) != 'Mn'
    ).lower().strip()

def word_overlap_ratio(s1, s2):
    """Fraction of stimulus words present in transcription"""
    s1_words = set(normalize_text(s1).split())
    s2_words = set(normalize_text(s2).split())
    if not s1_words:
        return 0
    return len(s1_words & s2_words) / len(s1_words)

def score_utterance(stimulus, transcription):
    """
    Score 0-4 based on meaning-based rubric (Ortega, 2000):
    4 = Exact repetition
    3 = Full meaning preserved, minor changes ok
    2 = More than half of meaning preserved
    1 = About half of meaning, significant info missing
    0 = No response, gibberish, or only 1-2 words
    """
    transcription_clean = clean_text(transcription)
    stimulus_clean = clean_text(stimulus)

    if not transcription_clean:
        return 0

    # Exact match normalized for accents (score 4)
    if normalize_text(transcription_clean) == normalize_text(stimulus_clean):
        return 4

    # Word overlap + semantic similarity combined
    overlap = word_overlap_ratio(stimulus_clean, transcription_clean)
    emb_s = model.encode(stimulus_clean, convert_to_tensor=True)
    emb_t = model.encode(transcription_clean, convert_to_tensor=True)
    sim = float(util.cos_sim(emb_s, emb_t))

    if sim >= 0.92 or overlap >= 0.90:
        return 3
    elif sim >= 0.75 or overlap >= 0.70:
        return 2
    elif sim >= 0.45 or overlap >= 0.40:
        return 1
    else:
        return 0

print("Scoring functions ready")

```

Scoring functions ready

```

In [16]: wb = load_workbook(SCORING_EXCEL_PATH)

for sheet_name in wb.sheetnames[1:]: # Skip Info tab
    ws = wb[sheet_name]
    print(f"Scoring {sheet_name}...")

    for row_num in range(2, 32):

```

```

stimulus_raw = ws.cell(row=row_num, column=2).value
transcription = ws.cell(row=row_num, column=3).value

if stimulus_raw:
    stimulus = clean_stimulus(stimulus_raw)
    score = score_utterance(stimulus, transcription)
    ws.cell(row=row_num, column=4, value=score)

wb.save(OUTPUT_SCORING_PATH)
print(f"\nSaved to {OUTPUT_SCORING_PATH}")

```

Scoring 38001-1A...

Scoring 38002-2A...

Scoring 38004-2A...

Scoring 38006-2A...

Saved to C:\Users\kashi\OneDrive\Desktop\AutoEIT\autoeit\output\test2_output.xlsx

```

In [17]: wb_out = load_workbook(OUTPUT_SCORING_PATH)
ws = wb_out["38001-1A"]

print("Sample scores for 38001-1A:\n")
for row_num in range(2, 12):
    stimulus = clean_stimulus(ws.cell(row=row_num, column=2).value)
    transcription = ws.cell(row=row_num, column=3).value
    score = ws.cell(row=row_num, column=4).value
    print(f"Score {score} | S: {stimulus[:40]}")
    print(f"          | T: {str(transcription)[:40]}")
    print()

```

Sample scores for 38001-1A:

Score 4 | S: Quiero cortarme el pelo
| T: Quiero cortarme el pelo

Score 4 | S: El libro está en la mesa
| T: El libro está en la mesa

Score 4 | S: El carro lo tiene Pedro
| T: El carro lo tiene Pedro

Score 4 | S: El se ducha cada mañana
| T: El se ducha cada mañana

Score 3 | S: ¿Qué dice usted que va a hacer hoy?
| T: Que dices ustedes se que van a hacer hoy

Score 2 | S: Dudo que sepa manejar muy bien
| T: Dudo que sepa manajar bien

Score 3 | S: Las calles de esta ciudad son muy anchas
| T: Las calles de esta ciudad son muy anchas

Score 2 | S: Puede que llueva mañana todo el día
| T: Puede que lleva mañana todo el día

Score 2 | S: Las casas son muy bonitas pero caras
| T: Las casas son muy bonitas pero muy cadas

Score 4 | S: Me gustan las películas que acaban bien
| T: Me gustan las peliculas que acaban bien

Methodology & Approach

Scoring System

Implemented the meaning-based EIT rubric (Ortega, 2000) with scores 0-4:

- **4:** Exact or near-exact repetition (normalized for accents)
- **3:** Full meaning preserved, minor changes acceptable
- **2:** More than half of meaning preserved, some inexactness
- **1:** About half of meaning, significant information missing
- **0:** No response, gibberish, or only 1-2 words

Technical Approach

- Multilingual sentence embeddings (paraphrase-multilingual-MiniLM-L12-v2) used for semantic similarity scoring
- Word overlap ratio combined with semantic similarity for robustness
- Accent normalization applied before exact match check
- Special notation ([pause], [gibberish], xxx) stripped before scoring

Threshold Calibration

Thresholds were calibrated by comparing model output against the provided example transcriptions in the scoring sheet, then adjusted to better match the rubric criteria.

Known Limitations

- Semantic similarity models may not perfectly capture all meaning distinctions relevant to L2 Spanish assessment
- Edge cases like synonymous substitutions and grammar changes that don't affect meaning are difficult to handle automatically
- Human validation against gold standard scores would be needed for production use